CSE110 Fall 2025 Lab Exam - 02 [Wed 2:00pm] Tentative Solutions and Rubrics

In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

[SET-A]

SOLUTION

```java
import java.util.Scanner;

public class LuckyNumberCheck{
 public static void main(String[] args){
  Scanner sc= new Scanner(System.in);
  System.out.print("Enter a positive number: ");
  int InputNum= sc.nextInt();
  int LuckyNumber=0;
  int TempNum=InputNum;

  //first iteratively determine the divisor based on the number of digits.
  int divisor=1;
  while (TempNum >10){
    divisor *= 10;
    TempNum /= 10;
  }
  // extract even digits from left to right and alternatively perform addition and subtraction.
  TempNum= InputNum;
  int position=0;
  while(TempNum>0){
   int digit = TempNum / divisor;

   if( digit % 2 == 0 && position % 2 ==0){ //leftmost even digit at even position > add
     LuckyNumber += digit;
     position++;
   }
   else if( digit % 2 == 0 && position % 2 !=0){ //leftmost even digit at odd position > subtract
     LuckyNumber -= digit;
     position++;
   }
   TempNum = TempNum % divisor; // removing the leftmost digit.
   divisor = divisor / 10; // reducing the divisor by a factor of 10 in each iteration.
  }
  System.out.println("Lucky Number is: " + LuckyNumber);

 }}
```

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[SET-B]

SOLUTION

```java
import java.util.Scanner;

public class LuckyNumberCheck{
  public static void main(String[] args){
    Scanner sc= new Scanner(System.in);
    System.out.print("Enter a positive number: ");
    int InputNum= sc.nextInt();
    int LuckyNumber=0;
    int TempNum=InputNum;
    //first iteratively determine the divisor based on the number of digits.
    int divisor=1;
    while (TempNum >10){
      divisor *= 10;
      TempNum /= 10;
    }
    // extract odd digits from left to right and alternatively perform subtraction and addition.
    TempNum= InputNum;
    int position=0;
    while(TempNum>0){
     int digit = TempNum / divisor;
     if( digit % 2 != 0 && position % 2 ==0){ //leftmost odd digit at even position > subtract
       LuckyNumber -= digit;
       position++;
     }
     else if( digit % 2 != 0 && position % 2 !=0){ //leftmost odd digit at odd position > add
       LuckyNumber += digit;
       position++;
     }
     TempNum = TempNum % divisor; // removing the leftmost digit.
     divisor = divisor / 10; // reducing the divisor by a factor of 10 in each iteration.
    }
    System.out.println("Lucky Number is: " + LuckyNumber);
    }}
```

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RUBRIC for Set A and Set B

SL	Points To Meet	Marks [10]
1	Mentioning **import java.util.Scanner** and Creating Scanner object	**0.5**
2	Java syntaxes (i.e. class, static void main, semicolon, curly braces and parentheses)	**0.5**
3	Taking user input .	**1**
4	Calculating the divisor properly.	**2**
5	Iterating properly through each digit in the number.	**3**
6	Checking and calculating odd-even positions.	**2**
7	Properly calculating the Lucky Number and printing the output correctly. [The output must match the format given in the question.]	**1**
	Total	**10**